

Xudong Zhu

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My research examines how large language models represent concepts and how these representations evolve during generation and training. I focus on the linear structure of these representations and its role in understanding and steering model behavior, as well as how different training methods, such as SFT and RL, reshape these learned representations.

RESEARCH INTERESTS

Representation Learning · Interpretability · Training Dynamics
Representation & Behavior Steering · Agent Behavior Control

EDUCATION

The Ohio State University Ph.D. in Computer Science and Engineering, advised by Prof. Zhihui Zhu	2024 – Present Columbus, OH
University of Electronic Science and Technology of China B.S. in Computer Science · GPA 3.98 / 4.00	2020 – 2024 Chengdu, China

PUBLICATIONS

- AbsTopK: Rethinking Sparse Autoencoders for Bidirectional Features**
Xudong Zhu, Mohammad Mahdi Khalili, Zhihui Zhu
[ICLR 2026](#) · [Paper](#) · [Code](#) · [Project Page](#)
- From Emergence to Control: Probing and Modulating Self-Reflection in Language Models**
Xudong Zhu, Jiachen Jiang, Mohammad Mahdi Khalili, Zhihui Zhu
[TMLR 2026](#) · [Paper](#) · [Code](#) · [Project Page](#)
- FCDS: Fusing Constituency and Dependency Syntax into Document-Level Relation Extraction**
Xudong Zhu, Zhao Kang, Bei Hui
[LREC-COLING 2024](#) · [Paper](#) · [Code](#) · [Project Page](#)
- Alleviating Subgraph-Induced Oversmoothing in Link Prediction via Coarse Graining**
Xudong Zhu, Dong Hao, Ziqin Gao, Liming Pan
[Neurocomputing, 2025](#) · [Paper](#) · [Code](#) · [Project Page](#)

HONORS & RESEARCH SUPPORT

MMLS 2026 Travel Award , Midwest Machine Learning Symposium	2026
NAIRR Pilot Project NAIRR260106 (contributor), National AI Research Resource Pilot Geometry and Training Dynamics of Representations in Large Language Models	2026

OPEN SOURCE

LLMUsage · Local-first LLM subscription usage monitor for Hyprrland (OpenAI, GLM, Grok, Claude)	TypeScript
dotfiles · fish, Neovim, and OpenCode configuration managed with GNU Stow	Lua